

Future Prospects of Turning Processes Optimization Using Metaheuristics Methods

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Abstract—In today's industries, optimization of turning processes is one of vital problems which aim to increase competitiveness and product quality. In fact, the decision of optimal machining parameters is arbitrary and complex. Traditionally, the selections is heavily relies on trial and error methods which is tedious and unreliable. To overcome these problems, metaheuristics methods has been proposed over the last decade. The results on several parameters such as cutting speed, depth of cut and feed rate indicate more effective and efficient way in comparison to traditional methods. This paper presents current development and future prospects due to metaheuristics techniques for optimizing the parameters of turning processes.

Keywords; *Turning process; metaheuristics; genetic algorithm; particle swarm optimization; hybrid algorithm*

I. INTRODUCTION

Determination of optimal cutting parameters is one of the most important and complex work in process planning and economics of metal cutting operation. It need many aspects including machining and mathematical knowledge and requires experience [1], [2]. Studies on metaheuristics algorithms have shown that these methods can be efficiently used to eliminate most of difficulties of classical methods. There are many parameters in turning processes and many conflicting objectives should optimize simultaneously [1], [2], [3]. Moreover, the turning process parameters to be optimized are highly constrained and nonlinear.

The conventional methods such as linear programming which using calculus-based and gradient methods cannot solve a multi-objective optimization problem and finding the global optimum. More difficulties is found in multi pass turnings because many practical constraints have to be considered [4], [5], [7], [9], [10] whereas modern heuristic methods include artificial neural network, Lagrangian relaxation, simulated annealing are more robust in global searching [3], [10]. However, they are usually slow in convergence, require higher computing time and may risk being trapped at a local optimum which fails to give best solution.

The objective of this paper is to present brief review of metaheuristics methods in turning process optimization. The

description includes optimization of cutting parameters, objective functions, constraints and algorithm implementation.

II. GENETIC ALGORITHM

In workshops, cutting parameters and part material are selected from machining database or handbooks, but these sources are slow, actually starting values and are not the optimal values [1], [2], [4]. Genetic algorithm is a powerful search algorithm and efficient tool for optimization that is able to solve complex problems very quickly [5], [9], [10], [11]. It simultaneously evaluates many points in the search space and more likely to find the global solution of a given problem [11]. However, single objective optimization is not suitable to repair the optimal cutting conditions due to the complexity of the machining processes. Therefore, multi objective is more preferable for these problems.

Ramon et al. (2006) studied a multi-objective optimization problems based on micro-Genetic algorithm to optimize the cutting parameters in turning processes [1]. Their used criterion like specific cost, machining time, material removal rate and tool life, and used also several different, conflicting objectives that should be optimized. The cutting parameters used are cutting speed (V), feed (f) and depth of cut (a). There are two objective functions which are production rate and tool life. The production rate (Z) measured as the total time:

$$Z = Z_s + \frac{V}{M} \left(1 + \frac{ZTC}{C} \right) + Z_o \quad (1)$$

whereas Z_s (set-up time), ZTC (tool change time), Z_o (time during which the tool does not cut), V (volume of the removal metal), T (tool life), M (material removal rate). The second objective function is tool life (ϵ) which can be calculated as:

$$\epsilon = \frac{V}{M \cdot T} 100\% \quad (2)$$

For a particular cutting process, Z_s , ZTC , Z_o are considered constant values, so that the objectives are functions of T and M and they depend on the decision variables (V , f , a) which are considered as constraints. The

authors showed that a two-point crossover is better than multi-point crossover. The mutation should be kept very low because the high value destroys good individuals.

Yildiz and Ozturk (2006) proposed hybrid GA with refined design space for solving the turning optimization problems [2]. The model of machining cost used machining variables (speed, feed, depth of cut) as:

$$C_i = c_i v^{a_{i1}} f^{a_{i2}} d^{a_{i3}} \quad (3)$$

whereas C_i : machining cost, c_i : cost component coefficients, v : machining speed, f : machining feed, d : depth of cut, $i=1 \dots n$ and a_{i1} , a_{i2} , a_{i3} : are machining variable exponents. The constraints B_n in this research can be expressed as in:

$$B_n = b_n v_n^{a_{m1}} f_n^{a_{m2}} d_n^{a_{m3}} \quad (4)$$

whereas b_n : term coefficient of constraints, $n=1 \dots N$, $m=1 \dots M$. The initial refinement was able to improve traditional GA.

Srikanth and Kamala (2008) proposed a real coded genetic algorithm (RCGA) to find optimum cutting parameter and estimate the minimum surface rough in turning operation [3]. The method is more efficient and robust in comparison to conventional genetic algorithm (CGA). RCGA is also known as floating-point or real number representation [13]. Individuals with the largest probability are selected as parent for the next generation. The crossover operation is controlled by arithmetic crossover operator (AMXO). The objective function is:

$$R_a = \frac{1.0632 \cdot f^{1.0198} \cdot d^{0.0119} \cdot H^{0.5234} \cdot r^{0.1388}}{V^{0.229}} \quad (5)$$

The parameters cutting speed (V), feed (f), depth of cut (d) and material hardness (H) has boundaries. The result is more accurate in comparison to traditional binary coded GA.

Car et al. (2009) proposed hybrid GA and linear programming optimization for turning process optimization [4]. The objective function used to minimize production time model (t_i) is expressed as:

$$t_i = t_g + t_a + t_p \quad (6)$$

whereas t_g : main machining time, t_a : tool change time, t_p : time during which the tool do not cut. The model is fast and suitable to optimize machine parameters.

Ahmad and Haque (2001) studied the optimization process planning parameters for machining rotational component using a genetic algorithm optimization toolbox developed in Matlab [5]. The constraints used are machining capacity such as feed rate, depth of cut and cutting speed.

The objective function i.e. machining time T is calculated as:

$$T_m = \frac{L_f n_{pass}}{f N_w} \quad (7)$$

whereas T_m : machining time, L_f : length of surface, n : number of passes, f : feed rate, N_w : rotational speed of work piece. The constraints used are cutting velocity, feed rate, depth of cut, rotational velocity of spindle and the power of machine. The results show that if the depths of cut and cutting speed are decreased while feed rate is increased, then the total machining time is optimum.

Recently, Shutong and Guo (2011) studied the minimization of the production cost in multi-pass turnings by using combined genetic algorithm with a pass repeating method [10]. This algorithm is different from other algorithms by putting the pass repeating process outside of the heuristic algorithms. Hence, the problem is turned into several easy sub-programs and become easier to find the optimal results. The optimal machining parameters to be optimized are cutting speed, feed rate and depth of cut for both rough and finish machining. The objective function used is:

$$UC = C_M + C_I + C_R + C_T \quad (8)$$

whereas UC : unit production cost, C_M : cutting cost by actual time in cut, C_I : machine idle cost for loading and unloading, C_R : tool replacement cost, C_T : tool cost. The practical constraints used are parameter bounds, tool-life, surface finish, force, power and roughing and finishing parameters relations. The numbers of possible value of rough cuts are calculated as (9):

$$m = (n_u - n_l + 1) \quad (9)$$

whereas m : number of rough cuts (sub-programs), n_u : upper limit, n_l : lower limit. The author proposed bound adjustment of optimized variants (BAOV) method to reduce the number of infeasible individuals in the iteration process and different from the chromosome encoding method in traditional GA. The crossover operator performs single-point between pairs of individuals, and the mutation operator performs inversion of binary bit string. The results show that this method was able to reduce unit production cost significantly. However, no real world application has been done.

Sanjeev et al. (2012) studied the influence of cutting parameters on the surface finish in turning poly tetra fluoro ethylene as an important class of material on CNC lathe [11]. Genetic algorithm is used to optimize the machining parameters. The objective function used to minimize the surface roughness is shown as:

$$R_a = -0.309 + 0.675V + 0.870f + 0.175d - 0.234V \cdot f - 0.002fd - 0.143V \cdot d \quad (10)$$

Whereas R_a : surface roughness, V : cutting speed, f : feed rate, d : depth of cut. The constraints considered in the function are v , f , and d . The single point crossover is set

with probability of 0.08. The mutation probability is set as 0.1 to prevent local convergence. However, many of the proposed approaches that using GA impose high computational time and can hardly find the global optimum.

III. PARTICLE SWARM OPTIMIZATION

Particle swarm optimization (PSO) algorithm has also been proposed in turning processes optimization [6], [10]. PSO is inspired by the social interaction between animal such as birds. Each particle (bird) update their position based on known best position (personal best) and the known best individual among the group (global best) [6], [7], [13]. The movement rule for updating the swam position is shown in (11) and (12):

$$V_i^{k+1} = w \cdot v_i^k + c_1 \cdot \text{rand}(p_{\text{best}i} - x_i^k) + c_2 \cdot \text{rand}(g_{\text{best}i} - x_i^k) \quad (11)$$

$$X_i^{k+1} = x_i^k + v_i^{k+1} \quad (12)$$

whereas v_i^k : velocity of particle i at iteration k , x_i^k : current position of particle i at iteration k , $p_{\text{best}i}$: personal best of particle i , $g_{\text{best}i}$: best position in the neighborhood, rand: random number between 0 and 1, w : weighting function and c_1, c_2 : learning rate.

Yigit and Ozel (2005) studied multi-objective optimization of hard turning using a neural network (NN) to model and predict process outputs such as the surface roughness and tool wear [6]. Then, PSO is used to obtain optimum process parameters of cutting tool wear, surface roughness and the productivity at the same time. The authors showed that to solve multi objective problems can be use two approaches. The first is by combining several objectives into a single objective through the use of “weights” or utility functions. The weakness of this way is largely depends on numerical weights that are difficult to select and not practical. The second way is by producing a Pareto-optimal set of non-dominated decision variable settings. This way avoids the problem in first method but suffers due to numerous prospective solutions. The objective function used in this work is to minimize the tool wear less than the limit value as shown in (13):

$$\min f = w_1 \times (\min(0, VB_{\text{lim}} - VB_{\text{pred}}))^2 + w_2 \times \left(\sum_{l=1}^{l_2} \min(0, Ra_{\text{lim}} - Ra_{\text{pred}}) \right)^2 + w_3 \times \frac{1}{x_1 x_2} \quad (13)$$

whereas VB_{lim} : limit of tool flank wear after given cutting distance, VB_{pred} : predicted tool wear at that point by neural network model, Ra_{lim} : limited surface roughness, Ra_{pred} : prediction of neural network model. The third term in the objective function represents productivity and has a penalty coefficient w_3 . The authors showed that if the neural network predictions are greater than limit values the results will have negative values therefore, minimum function in

the objective function will penalize these solutions by multiplying them with penalty coefficients w_1 and w_2 .

Tamizharasan et. al. [7] studied an optimal setting of product and process parameters in turning for minimizing the flank wear of tool and for improving the surface finish of the manufactured component. The parameters used are material hardness, cutting speed, feed rate, and depth of cut. The flank wear is selected as performance quality characteristic and the surface roughness as constraint. The machining duration is 300 seconds in order to avoid chipping of tool. At the end of each experiment the flank wear values of the cutting insert are measured and recorded using a tool maker’s microscope and the surface roughness measured using surface roughness tester. The statistical measure of quality called S/N ratio is applied to determine the effect of product and process parameters on flank wear. The lower ratio is used in this study because of minimization of the objective as:

$$S/N \text{ ratio for flank wear} = -\frac{1}{N} 10 \log_{10} \sum y^{-2} \quad (14)$$

whereas n : number of experiments and $\bar{y}$: measured data. The authors concluded that the *S/N ratio of flank wear* value that is obtained using PSO and non linear regression model are better in comparison to linear regression model.

Cardi et al. [8] studied a Neural Network (NN) training using particle swarm optimization (PSO) to know the effects of chatter on the quality of a machined part in turning operations. The root mean square error (RMSE) over the 1500 generations is used in training. The work piece displacement was predicted with an average RMSE of 1.41 and 1.70 mm for the two testing datasets. The method produced more accuracy level and more suitable for the control system.

Yigit and Tugrul (2005) studied the multi-objective particle swarm optimization (MOPSO) methodology to optimize machining parameters in hard turning process [6]. The relationships between machining parameters and the performance measures are obtained by using experimental data and swarm intelligent neural network systems (SINNS). This methodology is used in computing Pareto optimal non-dominated solutions where the neural networks are used in modeling the effects of decision variables on the performance measure of interest. The objective of the dynamic neighborhood-particle swarm optimization (DN-PSO) is used also which mean that instead of defining one global best for the whole population, neighborhoods are defined for each particle and gbests for each particle are found within these neighborhoods. Two case studies were taken; the first one is for maximizing the cutting tool life as in (15):

$$T = \frac{A}{V^3 + B \times V^2 + C \times V} \quad (15)$$

whereas T : tool life time, V : cutting velocity, A, B, C : calculated experimentally. The second case study for minimizing surface roughness and machining time as in (16):

$$Mrr = v \times f \times d \quad (16)$$

whereas Mrr : material removal rate, v : cutting velocity, f : feed rate and d : depth of cut. The constraints used are v, f , and d . The authors posit that decreasing the feed rate yields good surface finish.

Ganesan et al. (2011) studied the optimal machining parameters in turning process for machining a continuous finished profile and determines the minimum production time [9]. Cutting force, power, dimensional accuracy and surface finish were taken as constraints. The authors used a genetic algorithm and particle swarm optimization to solve the problem. The objective function used is to find minimum operation time as:

$$\begin{aligned} t_p &= T_m + t_e \frac{T_m}{t_l} + T_i \\ &= L_j + \frac{3}{f \cdot n} + t_e \frac{L_j + \frac{3}{f \cdot n}}{f \cdot d \cdot n} + T_i \end{aligned} \quad (17)$$

whereas t_p : unit time per work piece (min), T_m : cutting/machining time, t_e : time required to exchange tool (min), t_l : tool life (min), T_i : machine idle time, L_j : length of the travel, f : feed, n : speed, d : depth of cut. The crossover is done with a crossover probability of 0.75 and a single point crossover is used in this work. Mutation is a random modification of a randomly selected string and done with a mutation probability of 0.1. The results showed that PSO produces minimum fitness and gives better results and less computational time.

IV. CONCLUSION

The metaheuristics methods have been showed promising for turning process optimization. These methods are able to determine the machining process parameters more efficiently than non-metaheuristic methods. The comparison between GA and PSO shows that PSO is better than GA for non moving processes, easy to implement and more suitable. Moreover, GA requires high computational time and less effective. The other advantage of PSO is the ability to exchange information rapidly between particles and has better accuracy than GA. However, PSO algorithms can also trapped into local optimum which leads to premature convergence.

On the other hand, less effort has been done in the other metaheuristics algorithms such as artificial immune system

and artificial bee colony for turning processes optimization. In future, we aim to study these possibilities in more detail.

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